

Executor Framework Exercises

1. You are given a folder containing 100 text files. Each file has multiple lines of text. Implement a program using the Executor Framework that:
 1. Uses a fixed thread pool of size 5.
 2. Submits a Callable task for each file.
 3. Each task reads its file and returns the number of lines inside it.
 4. Collect and combine all results to compute the total number of lines across all files.
 5. Print the thread name processing each file and the total number of lines at the end.
 6. Ensure proper shutdown of the ExecutorService.
2. Create a system health monitor using ScheduledExecutorService that:
 1. Prints available processors, free memory, used memory, and the current thread's name every 2 seconds.
 2. Stops automatically after 20 seconds.
 3. Uses `scheduleAtFixedRate()` for periodic execution and `schedule()` to stop the monitor.